

Problem Set: Set Theory I

1. Identify whether each of the following indicators is measured at a nominal, ordinal, interval, or ratio level. Note also whether each is a discrete or a continuous measure:
 - (a) Highest level of education as (1) some high school, (2) high school graduate, (3) some college, (4) college graduate, (5) postgraduate.
 - (b) Annual income.
 - (c) State welfare expenditures, measured in millions of dollar.
 - (d) Vote choice among Bush, Clinton, and Perot.
 - (e) Absence or presence of a militarized interstate dispute.
 - (f) Military personnel, measured in 1,000s of persons.
 - (g) The number of wars in which countries have participated.
2. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$ and $C = \{6, 8\}$. Find following:
 - (a) $A \cup B$
 - (b) $A \cap B$
 - (c) $A \cap B^C$
 - (d) $B - A$
 - (e) $C - B$
 - (f) $A \cap C$
3. Let $A = \{1, 5, 10\}$, $B = \{1, 2, \dots, 10\}$.
 - (a) Is $A \subset B$, $B \subset A$, both or neither?
 - (b) What is $A \cup B$?
 - (c) What is $A \cap B$?
 - (d) Partition B into two sets, A and everything else. Call everything else C. What is C?
 - (e) What is $A \cup C$?
 - (f) What is $A \cap C$?

4. Let $A = \{a, b, c, d\}$, $B = \{1, 2, 3, 4\}$ and $C = \{a, b, 1, 2\}$. Show that:
 - (a) Distributivity: $(A \cap C) \cup (B \cap C) = (A \cup B) \cap C$
 - (b) Associativity: $(A \cap B) \cap C = A \cap (B \cap C)$
 - (c) De Morgan Laws: $C - (A \cup B) = (C - A) \cap (C - B)$
5. Determine which of the following formulas are true. If any formula is false, find a counterexample to demonstrate this using a Venn diagram.
 - (a) $A \setminus B = B \setminus A$
 - (b) $A \subseteq B \iff A \cap B = A$
 - (c) $A \cup B = A \cup C \implies B = C$
 - (d) $A \subseteq B \iff A \cup B = B$
 - (e) $A \cap B = A \cap C \implies B = C$
 - (f) $A \setminus (B \setminus C) = (A \setminus B) \setminus C$
6. Explain in words why it is true that for any sets A, B, C :
 - (a) $(A \cup B) \cup C = A \cup (B \cup C)$
 - (b) $(A \cap B) \cap C = A \cap (B \cap C)$
 - (c) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 - (d) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
7. Find the interior point(s) and the boundary points(s) of the set $\{x : 1 \leq x \leq 5\}$.
8. Why does every set in $\mathbb{R}$ that is nonempty, closed, and bounded have a greatest member?
9. Which of the following sets are open, closed, or neither?
 - (a) $D = \{x \in \mathbb{R}^1 : x = 2 \text{ or } 3 < x < 4\}$
 - (b) $A = \{(x, y) \in \mathbb{R}^2 : x^2 \leq y \leq 1\}$
 - (c) $B = \{(x, y) \in \mathbb{R}^2 : x^2 < y < 1\}$
 - (d) $C = \{(x, y) \in \mathbb{R}^2 : x^2 \leq y < 1\}$
 - (e) universal set
10. Sketch the following functions:
 - (a) $f(x) = 2$
 - (b) $f(x) = 3x - 1$
 - (c) $f(x) = x^2 + 2x + 1$
 - (d) $f(x) = (x - 3)^{-1}$
 - (e) $f(x) = |2x - 2|$

- (f) $f(x) = e^{2x}$
- (g) $f(x) = -\sqrt{x}$

11. Which of the following functions is injective, bijective, or surjective?

- (a) $a(x) = 2x + 1$
- (b) $b(x) = x^2$
- (c) $c(x) = \ln x$
- (d) $d(x) = e^x$

12. Simplify the following expressions as much as possible:

- (a) $(-x^4y^2)^2$
- (b) $9(3^0)$
- (c) $(2a^2)(4a^2)$
- (d) x^4/x^3
- (e) $(-2)^{(7-4)}$
- (f) $(1/(27b^3))^{(1/3)}$
- (g) $y^7y^6y^5y^4$
- (h) $((2a/7b)/(11b/5a))$
- (i) $(z^2)^4$

13. Solve:

- (a) $\sqrt[3]{2^3}$
- (b) $\sqrt[3]{2^7}$
- (c) $\sqrt[4]{625}$

14. Solve the following inequalities so that the variable is the only term on the left-hand side:

- (a) $x - 3 < 2x + 15$
- (b) $11 - (4/3)t > 3$
- (c) $(5/6)y + 3(y - 1) \leq (11/6)(1 - y) + 2y$

15. Find the roots (solutions) to the following quadratic equations:

- (a) $4x^2 - 1 = 17$
- (b) $9x^2 - 3x - 12 = 0$
- (c) $x^2 - 2x - 16 = 0$
- (d) $6x^2 - 6x - 6 = 0$
- (e) $5 + 11x = -3x^2$